

Credit Card Approval Prediction System

Abstract

This project predicts whether a credit card application will be approved or rejected using Machine Learning and Artificial Intelligence. It helps banks automate the credit approval process by analyzing applicant details such as income, employment, education, and family information. The application is developed using Python, Scikit-Learn, Flask, and HTML/CSS

Problem Statement

Banks receive thousands of credit card applications every day. Manual verification is time-consuming and error-prone. This project automates the approval process using machine learning, reducing processing time and improving decision-making.

Features

The Credit Card Approval Prediction System includes several important features that make the application efficient and user-friendly. It provides a simple web interface where users can enter applicant details such as income, education, employment status, and family information. The system uses a trained machine learning model to instantly predict whether the credit card application is likely to be approved or rejected.

The application includes an interactive dashboard that displays project information and prediction statistics. It performs data preprocessing and feature encoding to ensure accurate model predictions. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, were evaluated during model development to identify the best-performing model.

The project also supports real-time prediction through a Flask-based web application, allowing users to submit applicant details and receive immediate results. The interface is responsive, easy to navigate, and suitable for demonstration as well as educational purposes. Additionally, the project is

designed with a modular structure, making it easier to maintain, extend, and deploy on cloud platforms in the future.

Installation Step

1. Clone the repository.
2. Open the project in VS Code.
3. Create a virtual environment.
4. Install dependencies.

```
pip install -r requirements.txt
```

5. Run the application.

```
python app.py
```

6. Open

Usage Instructions

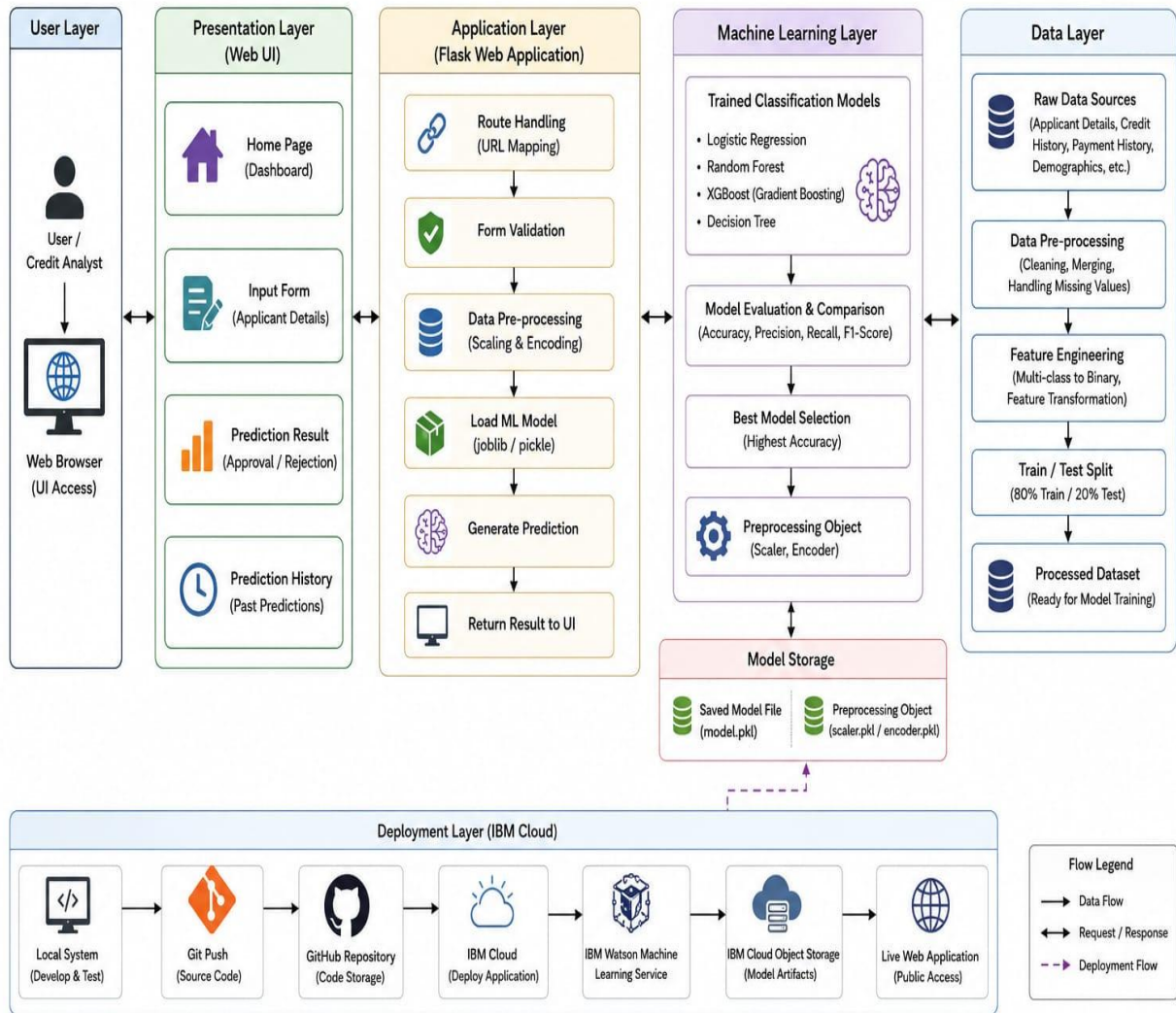
To use the Credit Card Approval Prediction System, first run the Flask application by executing the app.py file. Once the server starts successfully, open a web browser and navigate to <http://127.0.0.1:5000>. The home page of the application will be displayed, where users can access the dashboard or start the prediction process.

Click on the Start Prediction button to open the prediction form. Enter the required applicant details, including personal information, income details, employment information, education level, family status, and other financial attributes. After filling in all the required fields, click the Predict Credit Card Approval button.

The application processes the entered data using the trained machine learning model and instantly displays the prediction result, indicating whether the credit card application is Approved or Rejected. Users can return to the prediction page to test different applicant details or access the dashboard to view

project information and statistics. The application is designed to provide a simple, fast, and user-friendly experience for demonstrating machine learning-based credit card approval prediction.

Technical Architecture – Credit Card Approval Prediction Using IBM Watson Machine Learning



Screenshots

The documentation includes screenshots of the Home Page, Dashboard, Prediction Form, Approved Result page, and Rejected Result page. These screenshots demonstrate the complete working flow of the application from data input to prediction output.



Credit Card Approval Prediction System

Credit Card Approval Prediction



Go To Dashboard



Start Prediction

777,715

Total Applications

766,140

Approved

11,575

Rejected



Project Team Members

k.haripriya
Team leader

gonigunta shalusha
Team Member

H. J. Gnana Prasuna
Team Member

Gonigunta sailaja
Team Member

Gender

Male

Own Car

Yes

Own Property

Yes

Number of Children

3

Annual Income

500000

Income Type

Commercial associate

Education Type

Academic degree

Family Status

Civil marriage

Housing Type

Co-op apartment

Applicant ID

45673

Mobile Available

Yes

Work Phone

Yes

Phone

Yes

Email

Yes

Age (Days)

-12000

Employment (Days)

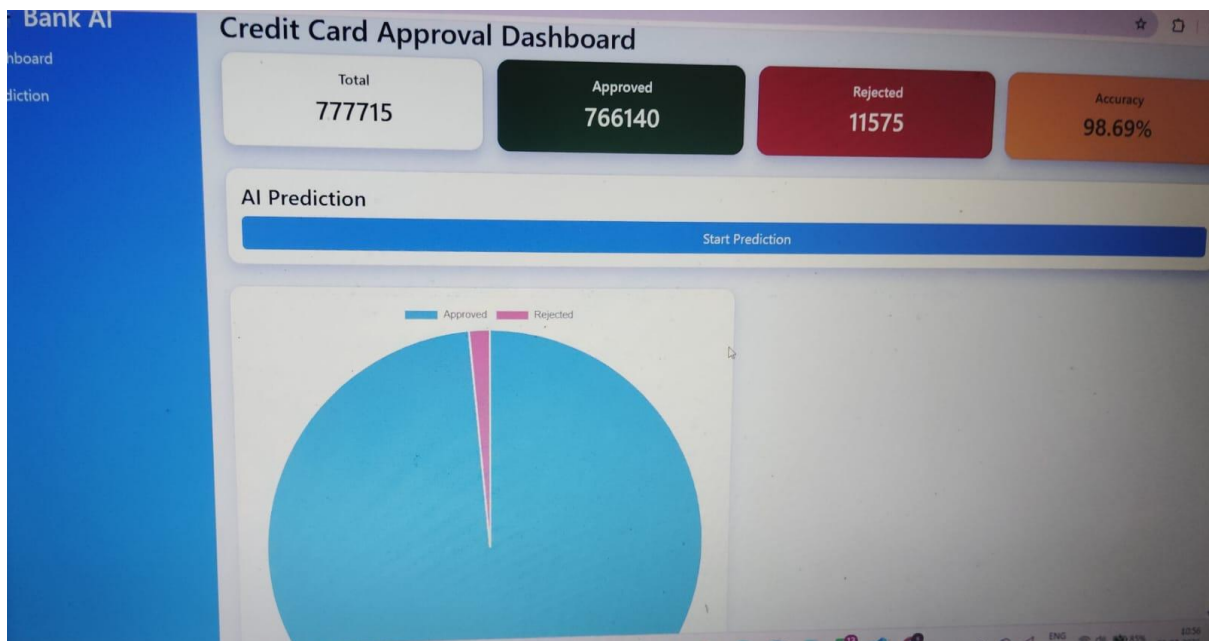
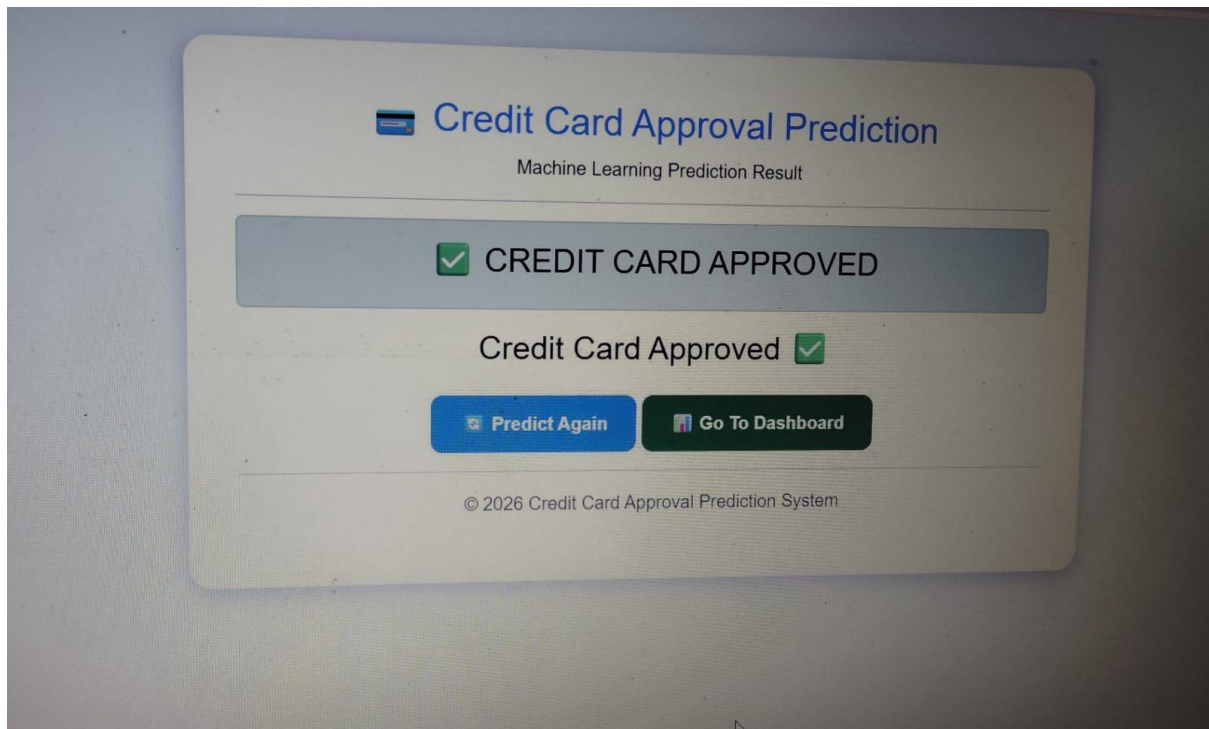
-3000

Occupation

Accountants

Family Members

5



Future Scope

The project can be enhanced by integrating more financial parameters such as credit score, loan history, EMI payment records, and banking transactions. Future improvements include deploying the application on IBM Cloud or AWS for real-time access, integrating a secure user authentication system, improving model accuracy using advanced machine learning techniques, and developing a mobile application for easier access.

Team Members

Team Leader: Hari Priya Koppa

Team Members:

Gonigunta shalusha

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H. J. Gnana Prasuna

Conclusion

The Credit Card Approval Prediction System successfully demonstrates how machine learning can automate the credit card approval process by analyzing applicant information and predicting whether an application is likely to be approved or rejected. The project reduces manual effort, improves decision-making efficiency, and provides quick prediction results through a user-friendly Flask web application. By combining data preprocessing, machine learning algorithms, and a responsive web interface, the system offers a practical solution for supporting credit card approval decisions. This project also provides a strong foundation for future enhancements, such as incorporating additional financial features, improving model.